Proof

By property 7, we have

$$\int_0^{2a} f(x)\,dx = \int_0^{a} [f(x) + f(2a-x)]\,dx. \quad \text{... (1)}$$

Setting the condition $f(2a-x) = -f(x)$ in equation (1), we get

$$\int_0^{2a} f(x)\,dx = \int_0^{a} [f(x) - f(x)]\,dx = 0.$$

■

Property 12

$$\int_0^a x\,f(x)\,dx = \frac{a}{2}\int_0^a f(x)\,dx \text{ if } f(a-x) = f(x).$$

Proof

$$\text{Let } I = \int_0^a x\,f(x)\,dx \quad \text{... (1)}$$

$$\text{Then } I = \int_0^a (a-x)f(a-x)\,dx, \text{ since } \int_0^a g(x)\,dx = \int_0^a g(a-x)\,dx$$

$$= \int_0^a (a-x)f(x)\,dx, \text{ since } f(a-x) = f(x).$$

$$\therefore I = \int_0^a (a-x)f(x)\,dx \quad \text{... (2)}$$

Adding (1) and (2), we get

$$2I = \int_0^a (x + a - x)f(x)\,dx$$

$$= a\int_0^a f(x)\,dx.$$

$$\therefore I = \frac{a}{2}\int_0^a f(x)\,dx.$$

■

Note

This property help us to remove the factor x present in the integrand of the LHS.

Example 9.20

Show that $\int_0^{\pi} g(\sin x)dx = 2\int_0^{\frac{\pi}{2}} g(\sin x)dx$, where $g(\sin x)$ is a function of $\sin x$.

Solution

We know that

$$\int_0^{2a} f(x)\,dx = 2\int_0^{a} f(x)\,dx \text{ if } f(2a-x) = f(x).$$

$$\text{Take } 2a = \pi \text{ and } f(x) = g(\sin x).$$

$$\text{Then, } f(2a-x) = g(\sin(\pi - x)) = g(\sin x) = f(x).$$

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